

Interface Specification: the on-chain and off-chain pieces

Companion to White Paper v2.0, Revision 4 (DOI 10.5281/zenodo.23087313) · Michael Twigg, PhD · ORCID 0009-0002-4607-6919

Ballots are counted from paper under state law; nothing is tallied on-chain. The chain holds only Vote Coin mints and burns (Phases I–II), Node-signed batch anchors and one registry record. **S** = in the white paper (section, page). **P** = proposed here, open for review; a **P** line that departs from the paper says so.

A Records and signatures · off-chain, built inside the air-gapped Secure Node

Ballot record R	UTF-8 lines, each ending in a line feed: INDEX: dddddd, then CONTEST: SELECTION for each contest in ballot order (NO SELECTION if blank), spelled as in the election definition.	S §8.1 p31
Index, salt, f	Index: 8 random digits, unique per Node per election, printed in OCR-B as dddd dddd, never on the receipt. Salt s: 32 random bytes (SP 800-90A/C), sealed log only. Fingerprint f = SHA-256(R s); the tablet's Vd must equal the scanner's Vp, or the paper governs; the receipt's tracking code is f.	S §8.1 p31 S §15.2 p62
Batch and root	Closes at 60 minutes or 4,096 ballots, never under 100; a short last batch joins the one before (it may pass 4,096), and a Node with under 100 in all anchors one batch. root = RFC 9162 §2.1.1 Merkle root (SHA-256) of the fingerprints, sorted ascending.	S §8.1 p31–32
Credential tag t	32 bytes per voter per election: t = HMAC-SHA-256(k_card, election_id) (FIPS 198-1). k_card, a 32-byte secret made at enrollment, sits in the card and, sealed, in the jurisdiction's issuance records, so the jurisdiction can mint t for each election and only it can link t to a voter. The card presents t at check-in; the §15.2 proof covers t.	S Glossary p84 P construction, escrow (departs Obj. 26 p79)
Tag root	tag_root = RFC 9162 root of the batch's spent tags, sorted the same way (every phase); with none, tag_count = 0 and tag_root = SHA-256 of the empty string (RFC 9162 §2.1.1). Both lists are sorted, so no position links a tag to a fingerprint.	P separate root (departs from §12.1 p46)
election_id	32 bytes: SHA-256 of a published UTF-8 descriptor, one field per line: jurisdiction, election date, election type, serial.	P
Batch header	Fixed 119 bytes, integers big-endian: version (1) election_id (32) node_id (2) seq (4) window_start (8, POSIX seconds, on the hour) ballot_count (4) tag_count (4) root (32) tag_root (32). Fixed offsets let a script read a field without parsing.	S §4.3 p16, §8.1 p32 P layout
Node signature	sig = Ed25519 (RFC 8032) over the 119-byte header: 64-byte signature, 32-byte key. ML-DSA (FIPS 204) can later sign anchors, split into 64-byte metadata chunks; CastBurn stays Ed25519 until Plutus adds a post-quantum builtin.	S §4.3 p16 P signed bytes, PQ path

B Vote Coin: token and minting policy · on-chain, Phases I–II

Asset	Cardano native asset: policy ID (28 bytes) + asset name = t (32 bytes, the maximum); one coin per voter.	S §14.1 p57 P name = t
Minting policy	Plutus V3, in Aiken. Fixed parameters: election_id, validator hash, registry token, enrollment_close, closeout_open (POSIX ms). Redeemer Issue CastBurn CloseOut. No other policy may mint or burn in the transaction. No output may hold a Vote Coin, except in Issue, whose outputs' Vote Coins equal the mint field exactly, one coin alone per output at the validator.	S Plutus §14.2 p59 P the rest
Issue (+1)	Every quantity +1. Spends an input locked by the Triad's 3-of-3 native script; validity ends by enrollment_close, before the first ballot. Each coin goes alone to the validator, with inline datum {election_id, this policy's ID} and a jurisdiction-funded deposit.	S §14.1 p57, §VI p27 P 3-key issue, time rule
CastBurn (−1)	Every quantity −1, after enrollment_close. Redeemer {header, sig, start, boundary}: the registry key for header.node_id verifies sig; election_id matches; tag_count ≤ ballot_count; the burned names, read from the mint field, are exactly a contiguous run of the sorted tags from start, proved to tag_root by one Merkle multiproof; all spent coins' ada goes to election_account.	S rule §4.2 p14 P mechanism
CloseOut (−1)	Every quantity −1. Spends a Triad-script input, signed only once certification and the challenge period are over; validity begins no earlier than closeout_open. Burns the unused coins, in as many transactions as it takes; the count is published.	S §14.1 p59 P mechanism
No re-issue	Minting ends at enrollment_close and burns begin after it, so no coin can be minted once any is spent. A tag minted twice before the close is not blocked by the script, but shows a supply of 2, which anyone can check.	S §15.4 p63 P mechanism

C Validator and registry · datum and redeemer shapes

Burn-only validator	Plutus V3 spending script; inline datum {election_id, policy ID}; redeemer Burn. Passes only if the policy named in its datum runs in the same transaction; that policy checks every coin once and lets no output hold one, so each coin spent is burned.	S burn-only §4.2 p14 P shapes; policy burns
Registry	One UTxO at the Triad's 3-of-3 native-script address (statutory alternates included), holding a one-shot token; the policy reads exactly one reference input (CIP-31) holding it and takes the Triad's script from its address. Inline datum (CIP-32): {election_id, node_keys [(node_id, Ed25519 key)], election_account, l_commit = hash of the §15.2 statement}; only that script can change it.	S 3-of-3 §VI p27 P structure
Deposits and fees	Every burn returns the deposits to election_account, never to a voter (52 U.S.C. § 10307(c)). Fees come from a small, capped float that the workstation's single key spends, a bounded departure from §4.1's multi-signature rule.	S §4.1–4.2 p14 P float (departs §4.1)

D Anchors · on-chain data, no script

Anchor transaction	Built and submitted by the anchoring workstation, outside the voting system; one transaction may carry many Nodes' batches, within the 16,384-byte limit. Final once more than 129,600 slots (36 hours) deep.	S §4.3 p16, §8.1 p32 S §12.1 p46–47
Metadata	Transaction metadata under one label registered in CIP-10. One entry per batch: the nine header fields and sig, each value 64 bytes or less (the metadata limit). Anyone can rebuild the header and check sig; an entry that fails is ignored.	P

E Public verification site · the off-chain index; nothing is tallied here

Files	registry.json (registry datum, its transaction) · n/{node}/b/{seq}.json (header, sig, both sorted lists, anchor and burn transactions, slot, final yes/no) · f/{fingerprint}.json (audit path to the root).	S lists §8.1 p32 P format
Receipt check	Tracking code f → audit path (side, hash) → root → anchor, exactly as Figure 16 shows for ballot 0515 0293.	S §12.1 p46, Fig. 16 p48
Anyone checks	Rebuild root and tag_root · verify sig with the registry key · match the anchor · one header per (node_id, seq), no gaps · every burn's header anchored · each tag in one batch, supply 1 · each Node's burns = its card ballots · minted = burned at the polls + at close-out.	S §4.3 p16, §8.1 p32 S §12.1 p46, §14.1 p59 S §15.4 p63, Obj. 18 p77 P list
Never published	Ballot content (R and vote packages), salts, validation indexes. Results come from the paper canvass, not the chain. Also private: k_card and the issuance records.	S Glossary p86, §8.1 p31 S §15.2 p62, Fig. 14 p40 S §12.1 p46 P k_card

Open before building: register the metadata label (CIP-10) · measure CastBurn bytes and execution units on testnet against the paper's estimate of about 100 bytes per burn (§12.1 p46) · audit both scripts independently · settle the §15.2 statement that binds t.